Relaxing Entry Constraints: College Expansion and Tertiary Attainment in Rural India

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Focus of the Paper

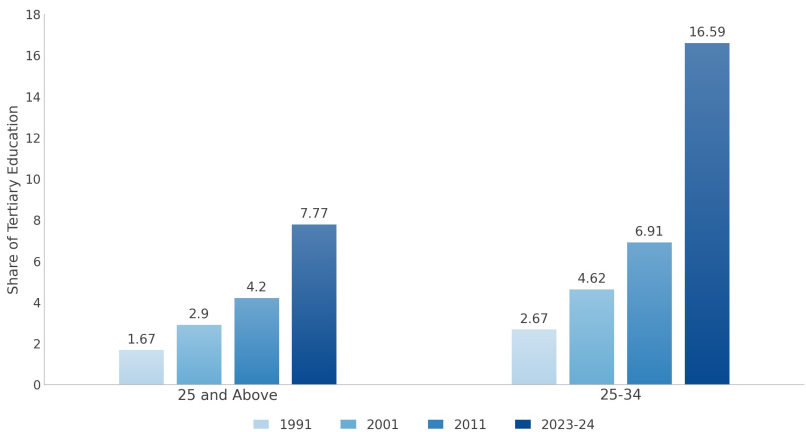
- Tertiary education raises worker productivity and promotes economic growth (Moretti, 2004a; Valero and Van Reenen, 2019).
  - Returns to higher education have increased over time (Psacharopoulos and Patrinos, 2018).
- In India, tertiary education contributes to economic growth and regional development (Castelló-Climent and Mukhopadhyay, 2013; Castelló-Climent et al., 2018).
- Yet, returns to skills are higher in cities, encouraging colleges to cluster in and around urban areas (Glaeser and Maré, 2001; Moretti, 2004b).
  - However, low rural-to-urban migration (Munshi and Rosenzweig, 2016) may limit rural youths' access to college education.

Focus of the Paper

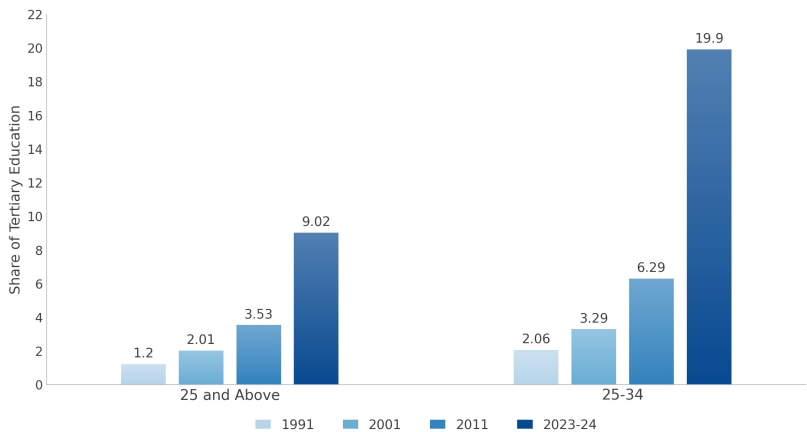
- Tertiary education has expanded worldwide, most rapidly in developing countries (Barro and Lee, 2013).
  - In India, tertiary attainment has risen sharply since the 1990s, particularly among rural populations and young adults.
- The number of colleges has increased alongside tertiary attainment.
  - Colleges are concentrated in towns, and rural-to-urban migration is limited. Why, then, has tertiary attainment among rural populations increased?
  - Colleges on the urban fringe may be accessible to nearby villages.
- Rajasthan follows India's trend in rural tertiary attainment but has experienced the fastest expansion in college numbers.
- I link geocoded college locations to census data on cohort- and village-level attainment and exploit staggered college openings across villages.
- Access to a college within 15 km raises tertiary attainment by $\approx$1.4 pp after two years and $\approx$2.2-2.4 pp after four to five years.

Motivation: Tertiary Education Attainment

- Share of the rural population with tertiary education:
  - India, 25 and above: 1.7% (1991) $\rightarrow$ 7.8% (2023-24).
  - India, 25-34: 2.7% $\rightarrow$ 16.6%.
  - Rajasthan, 25-34: 2.1% $\rightarrow$ 19.9%.



(a) India

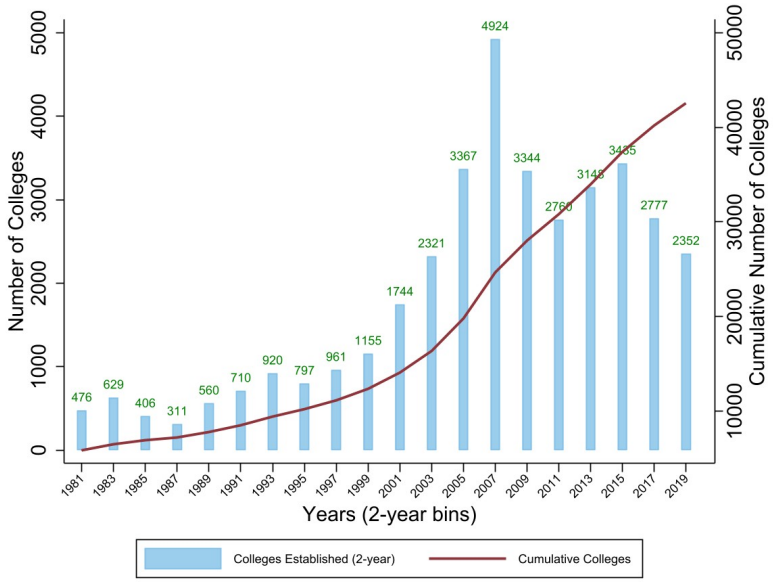


(b) Rajasthan

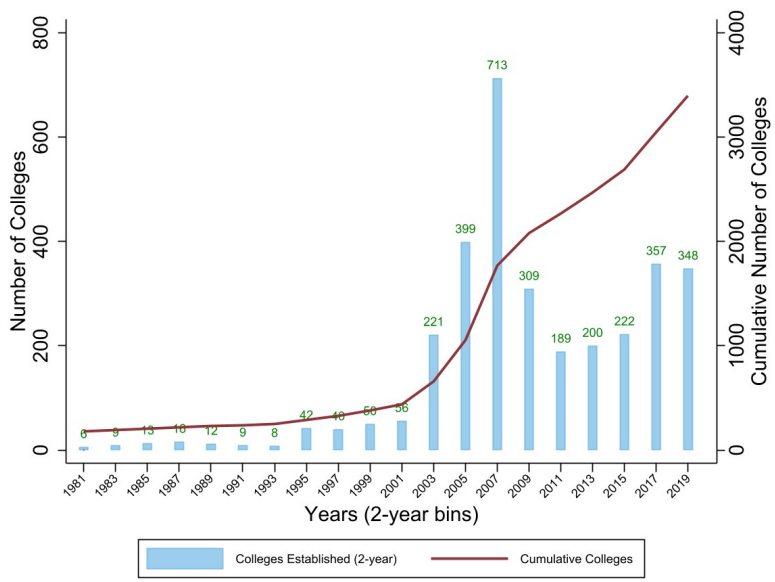
Source: Population Census (1991, 2001, 2011); PLFS (2023-24)

Motivation: Expansion of Colleges

- Colleges in India nearly tripled: $\approx$ 15,000 (2003) $\rightarrow \approx$ 43,000 (2020).
  - Accelerated after regulatory changes in the early 2000s opened higher education to private, self-financed institutions.
- Rajasthan is the outlier:
  - Fewer than 60 colleges opened in any two-year period before 2003; more than 1,300 opened in the six years after.
  - CAGR of 11.3% (2003-2020), nearly double the national 6.2% and the highest among states.



(a) India



(b) Rajasthan

Source: AISHE 2021-22.

Contextual Background

- Nationally, college establishment accelerated after 2002, following the UGC Private University Regulations (2003) and a planning push toward private provision.

- Rajasthan experienced the fastest expansion among Indian states:
  - The expansion was driven overwhelmingly by private entrants.

- Entry was governed by input-based requirements. Under the Rajasthan Private Universities Act, 2005, sponsors were required to have:
  - 30 acres of land and 10,000 m^2 of covered space (compared with 100 acres in Tamil Nadu);
  - INR 10 lakh in books and journals and INR 20 lakh in equipment and furniture;
  - an endowment of INR 2 crore, reduced to INR 1 crore away from major cities and INR 75 lakh in scheduled tribal areas.

Research Question

- Did expanded local college access increase tertiary attainment among the cohorts exposed to it?
  - To answer this question, I also ask: What determined the locations of new colleges, and did the predetermined structure of local landholdings shape where they opened?

Preview of Findings

- College Access and Tertiary Attainment:
  - A college within 15 km raised the tertiary share of exposed cohorts by $\approx$ 1.4 pp after two years and $\approx$ 2.2 to 2.4 pp after four to five years.
  - There is no evidence of pre-trends.
  - Robust to never-treated as a control group, 10-14 km bandwidths, a narrower cohort window, conditioning on local economic activity, and continuous treatment.
  - Both men and women gain. Women's effect rises steadily to $\approx$ 2.4-3.0 pp, matching or exceeding men's ($\approx$ 2.3-2.4 pp) by the end of the window.
- Land Fragmentation and College Placement:
  - Villages in districts with more fragmented landholdings in 1999 gained colleges faster: a 0.1 rise in fractionalization $\Rightarrow \approx$ 2.54 more colleges within 15 km (1999-2007).
  - The association was insignificant through 2001 and grew with the expansion, roughly sixfold by 2007. This is consistent with cheaper land for land-intensive private entry.

Contribution to the Literature

1. Educational supply and attainment:
   - Evidence on access to schools and colleges and its effects on educational attainment (Duflo, 2001; Burde and Linden, 2013; Muralidharan and Prakash, 2017; Khanna, 2023; Card, 1993; Currie and Moretti, 2003; Jagnani and Khanna, 2020; Sekhri et al., 2024).
   - → I study a large-scale, largely private college expansion at the village level using complete-enumeration census data.

2. Urban growth and the rural economy:
   - Growing towns matter for rural welfare (Christiaensen and Todo, 2014; Gibson et al., 2017).
   - → I show a new channel through which growing towns affect nearby villages: college establishment brings higher education within reach of rural residents.

3. Landholding structure and human capital:
   - Land distribution shapes agricultural productivity and investment in schooling (Adamopoulos and Restuccia, 2014; Foster and Rosenzweig, 2022; Galor et al., 2009; Banerjee and Iyer, 2005).
   - → I show that landholding structure can shape where colleges are established, in contrast to the state-led establishment of U.S. land-grant colleges (Kantor and Whalley, 2019; Andrews, 2023).

Data Sources

1. All India Survey on Higher Education (AISHE), 2021-22: Universe of colleges, including establishment year, type, and management.

2. Socio-Economic and Caste Census (SECC), 2011: Tertiary educational attainment by birth cohort in rural Rajasthan.

3. NSS 55th Round, 1999-2000: Household landholdings used to construct district-level land fractionalization.

4. Population Census, 2001: Scheduled Caste and Scheduled Tribe shares, population density, and non-farm employment.

5. Geographical variables: Constructed using ArcGIS.

Data: Constructing Access and Attainment

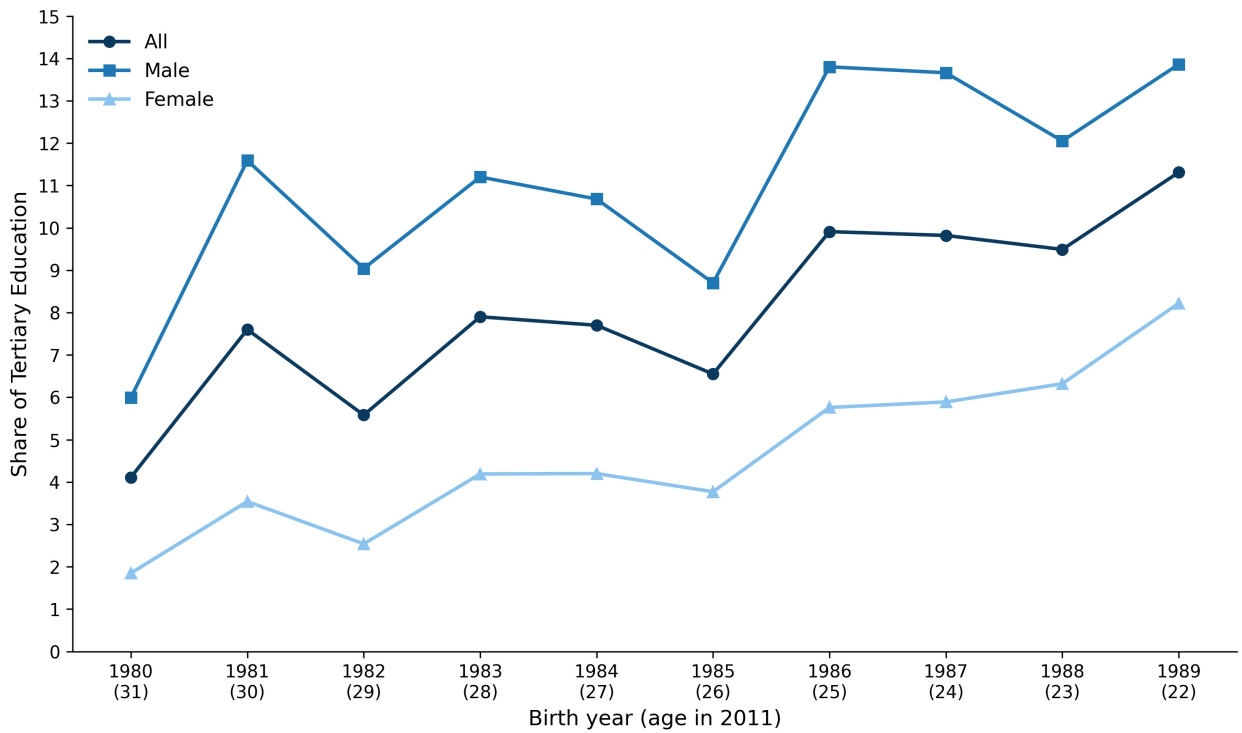
- College access (village v, year t)
  - College stocks are reconstructed using establishment years.
  - Number of colleges within 10-15 km of each village centroid, 1998-2007.
  - Binary measure: at least one college within b km. Continuous measure: number of colleges within b km.

- Cohort timing Cohort construction
  - Individuals aged 22-31 in 2011, corresponding to birth cohorts 1980-1989.
  - College-entry year $= 2011 -$ age in $2011 + 18 \in [1998, 2007]$.
  - Five cohorts entered college before the expansion (1998-2002) and five during the expansion (2003-2007).

- Outcome variable: Share of individuals in each village-cohort cell with a graduate degree or higher.

Tertiary Attainment by Birth Cohort and Gender

- Overall attainment: 4.1% for the 1980 cohort, rising to 11.3% for the 1989 cohort.
- Men's attainment: 6.0% to 13.9%; women's attainment: 1.9% to 8.2%.



Source: SECC 2011, rural Rajasthan, individuals aged 22-31.

Empirical Strategy: Staggered DiD

- Exposure occurs at different times across villages, and treatment effects may vary with exposure duration.

- First exposure: $G_v = \min\{t : N_{vt}(b) \geq 1\}$; $G_v = 0$ for never-exposed villages.

- Callaway and Sant'Anna (2021) estimator with doubly robust adjustment (Sant'Anna and Zhao, 2020) and not-yet-exposed controls:

$$ATT(g,t) = E[\, Y_t - Y_{g-1} \mid G_v = g \,] - E[\, Y_t - Y_{g-1} \mid G_v > t \,]$$

- Aggregate group-time effects by event time $e = t - g$:

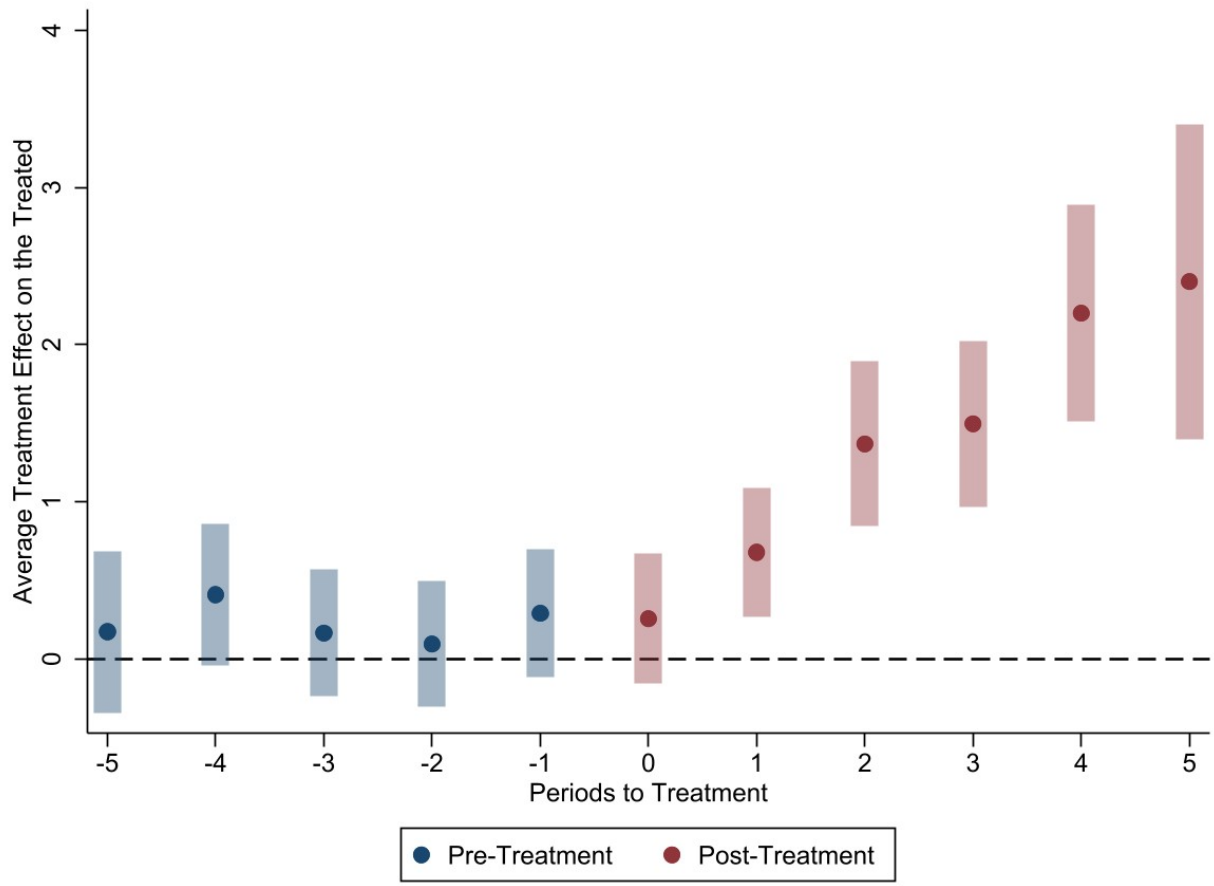
$$\theta(e) = \sum_g w_g(e)\, ATT(g, g+e),$$

  where $w_g(e)$ is the share of villages in cohort g observed at event time e.

- Cohorts that completed their college-going years before nearby colleges opened were never exposed; hence, the estimates of $\theta(e < 0)$ serve as a direct test of pre-trends.

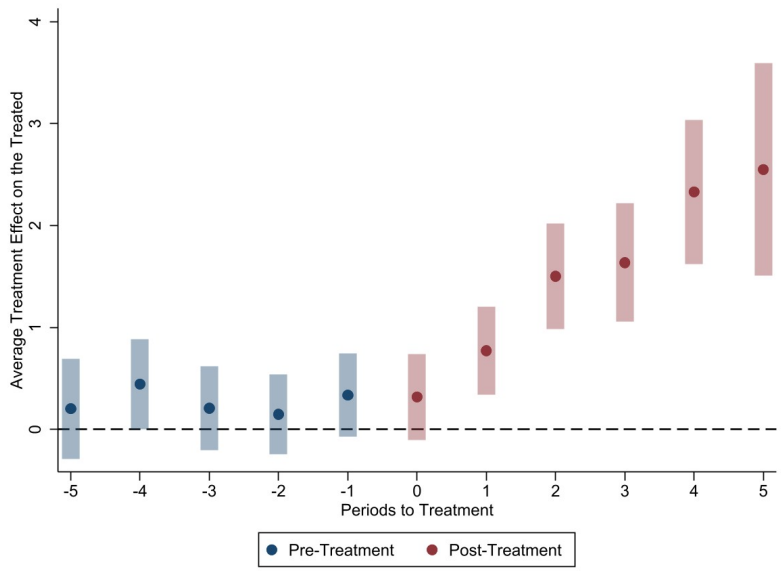
Main Result: 15 km Bandwidth

- Pre-exposure estimates are small (0.1-0.4 pp), individually insignificant, and show no trend.
- Post-exposure estimates are insignificant at $e = 0$, rise to approximately 0.7 pp at $e = 1$ and 1.4-1.5 pp at $e = 2, 3$, and reach 2.2-2.4 pp at $e = 4, 5$.

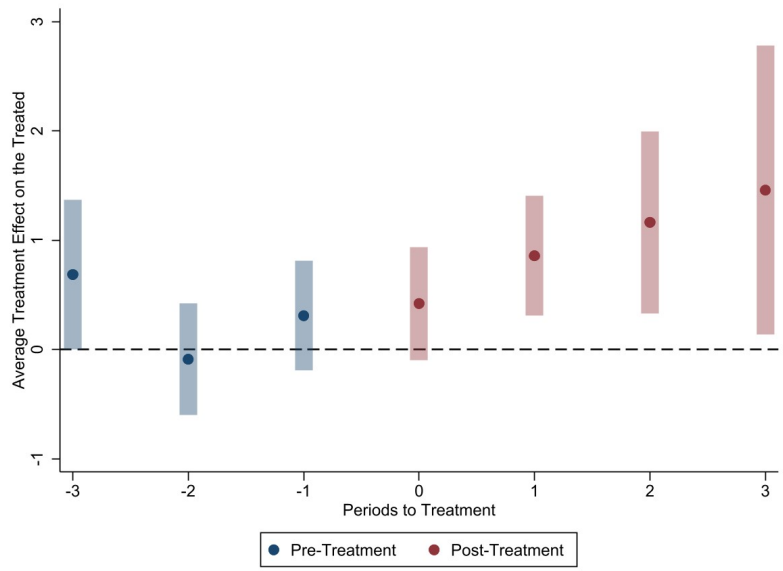


Robustness: Comparison Group and Cohort Window

- Never-treated comparison group: nearly identical estimates; $\approx$1.5-1.6 pp at $e = 2, 3$ and $\approx$2.3-2.5 pp at $e = 4, 5$.
- Cohorts 2000-2005: $\approx$0.9 pp at $e = 1$, 1.2 pp at $e = 2$, and 1.5 pp at $e = 3$, close to the baseline estimates.



(a) Never-treated comparison group

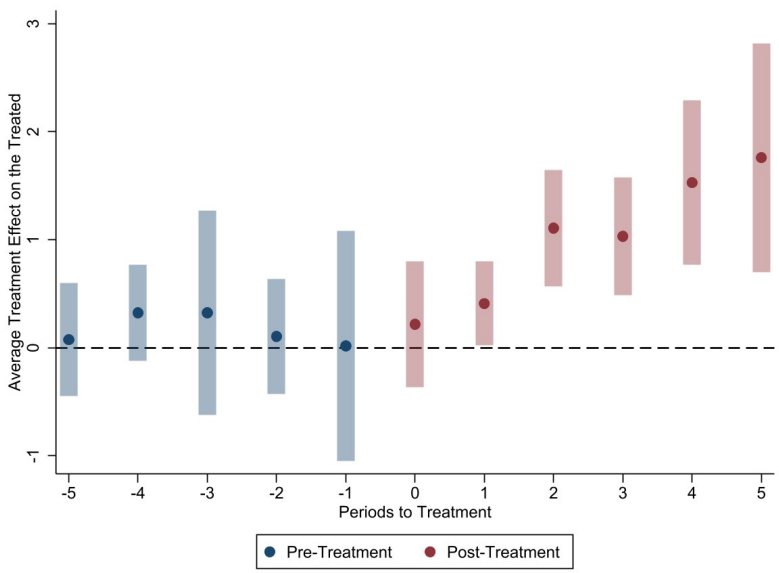


(b) College-entry cohorts, 2000-2005

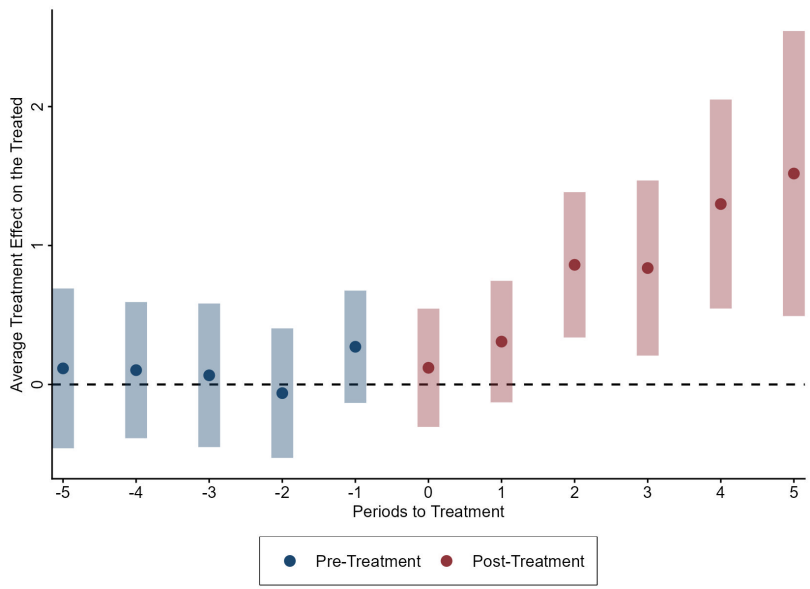
Robustness: Local Economic Activity

- Concern: colleges may have opened in villages that were already on different development paths. However, nightlights may also respond to college openings.

- Following Caetano et al. (2026):
  - Pre-treatment nightlights (at $g-1$): $\approx$1.1 pp at $e=2,3$; $\approx$1.5-1.8 pp at $e=4,5$.
  - Bad-control adjustment: effects are significant from $e=2$ ($\approx$0.9 pp) and reach $\approx$1.5 pp at $e=5$.



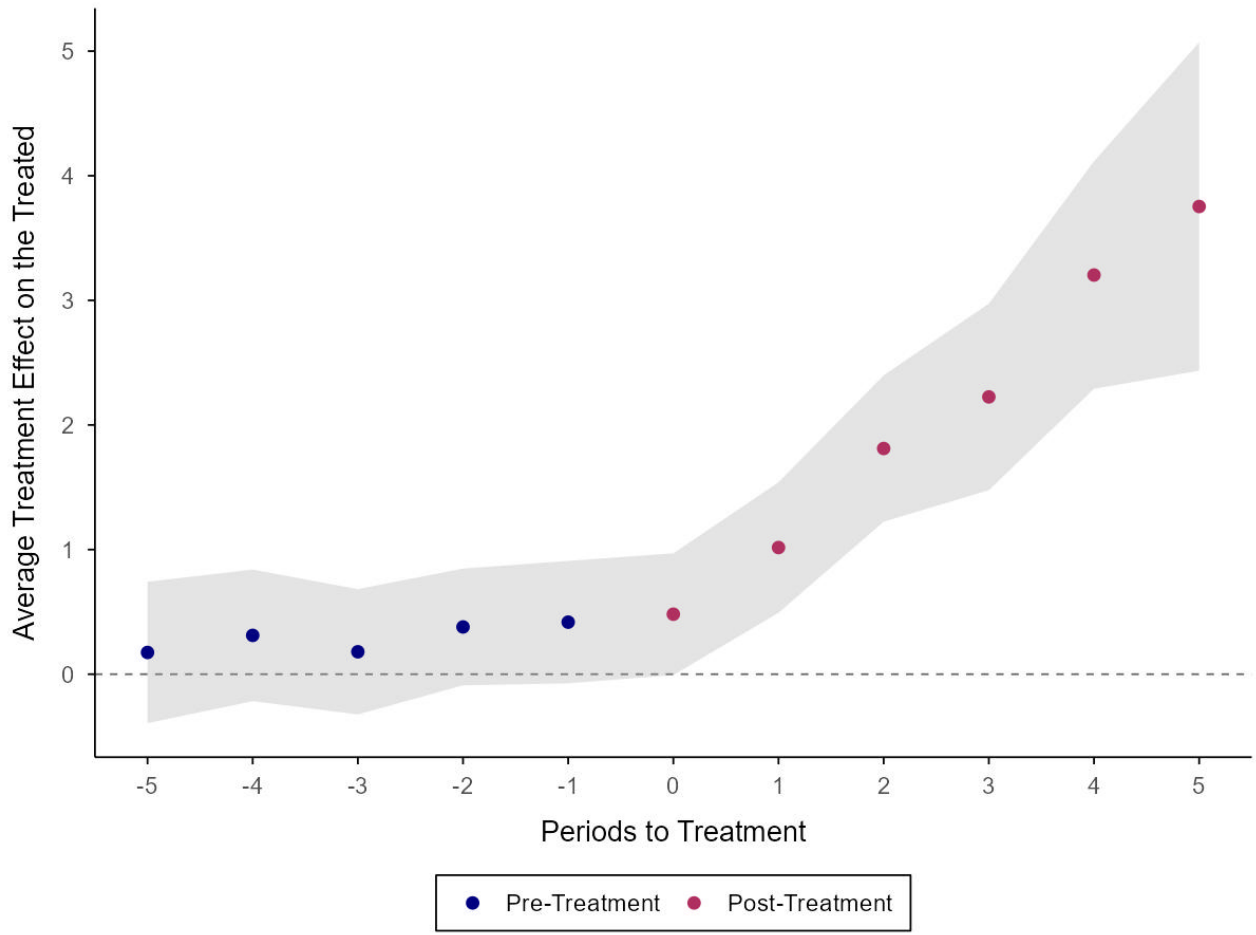
(a) Pre-treatment nightlights



(b) Bad-control adjustment

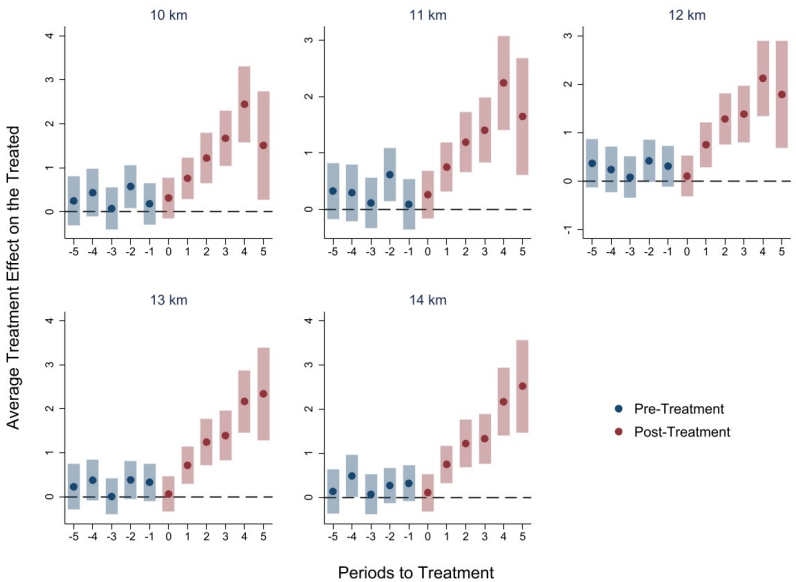
Robustness: Continuous Treatment

- Callaway, Goodman-Bacon and Sant'Anna (2024): dose $D_v = N_v(b) / \max_{v'} N_{v'}(b) \in [0, 1]$.
- The dynamics mirror the binary design: effects are flat before exposure. At the maximal dose, the estimated effects are ≈ 1.0 pp at $e = 1$, 2.2 pp at $e = 3$, and 3.8 pp at $e = 5$.

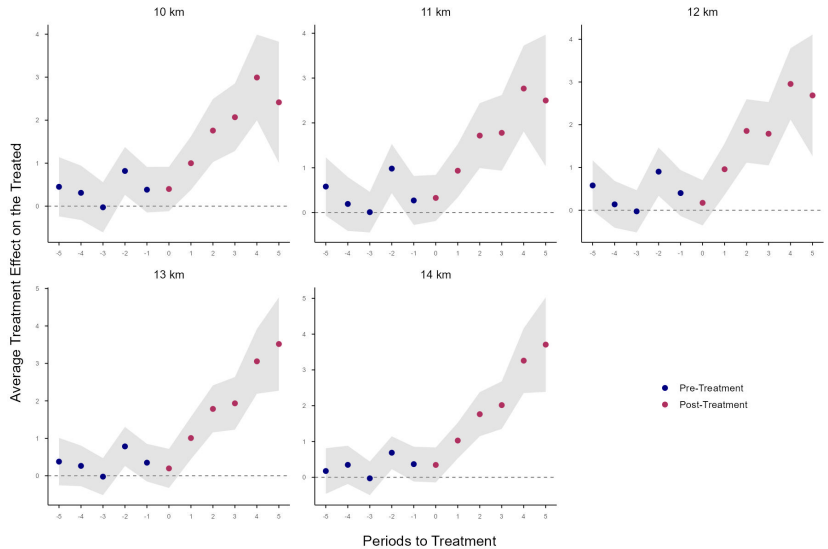


CS-DiD (10-14 km): Binary and Continuous

Binary treatment



Continuous treatment

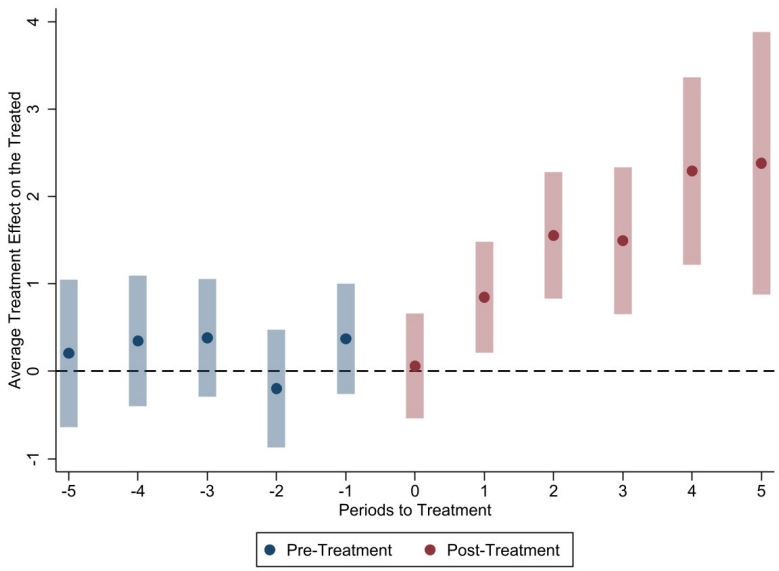


- Effects are positive from $e = 1$; $\approx$ 2.1-2.5 pp at $e = 4$ across all bandwidths.
- At $e = 5$, effects are smaller at 10-12 km (1.5-1.8 pp, with fewer adoption cohorts) than at 13-14 km (2.3-2.5 pp).

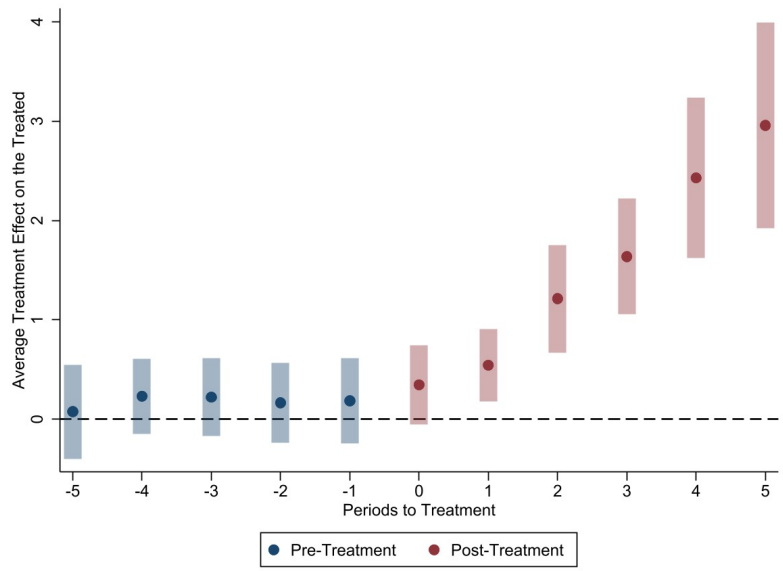
- Effects are positive from $e = 1$; $\approx$ 2.8-3.3 pp at $e = 4$ across all bandwidths.
- At $e = 5$, the pattern is similar to the binary design: 2.4-2.7 pp at 10-12 km and 3.5-3.7 pp at 13-14 km.

Heterogeneity by Gender

- Men: $\approx$0.8 pp at $e = 1$; $\approx$1.5 pp at $e = 2, 3$; $\approx$2.3-2.4 pp at $e = 4, 5$.
- Women: $\approx$0.5 pp at $e = 1$; $\approx$1.2 pp at $e = 2$; $\approx$1.6 pp at $e = 3$; $\approx$2.4-3.0 pp at $e = 4, 5$.



(a) Men



(b) Women

Land Fractionalization: Motivation and Measure

- Entry required a contiguous parcel of 30 acres.
- Land fragmentation creates operational constraints that can limit mechanization and lower yields.
- District land fractionalization, NSS 55th Round (1999-2000):

$$\mathrm{LF}_d = 1 - \sum_h s_{hd}^2, \qquad s_{hd} = \text{household } h\text{'s share of district rural land}$$

  - Near 0: land is concentrated among a few large holders.
  - Near 1: many households each hold a small parcel.
  - Measured before the post-2003 expansion.

Land Fractionalization: Specification

- Time-varying model with village FE and time-interacted baseline covariates:

$$C^b_{v,t} = \mu_v + \lambda D_t + \beta_{1999} F_{d(v)} + \beta_\Delta F_{d(v)} \times D_t + X_v' \delta_{1999} + (X_v \times D_t)' \delta_\Delta + u_{v,t}$$

  - $C^b_{v,t}$: colleges within b km of village v; $F_{d(v)}$: district fractionalization in 1999; $D_t = \mathbb{1}[t = 2007]$.
  - X_v: SC/ST shares, population density, elevation, ruggedness, sandy-soil share, cropland share, nighttime lights, and nonfarm employment.

- First-differencing eliminates μ_v and the time-invariant level terms. The specification also controls for the 1999 college stock:

$$\Delta C^b_v \equiv C^b_{v,2007} - C^b_{v,1999} = \alpha + \beta\, F_{d(v)} + \gamma\, C^b_{v,1999} + X_v' \delta + \varepsilon_v$$

  - Tests whether villages in more fragmented districts accumulated colleges faster, rather than whether they had more colleges initially.

Growth in Colleges, 1999 to 2007

- The coefficient is positive and statistically significant at the 5% level across all bandwidths, increasing from 13.07 (10 km) to 25.40 (15 km).
- At 15 km, a 0.1 increase in fractionalization is associated with approximately 2.54 additional colleges within the radius of the average village.

	Change in number of colleges, 1999-2007					
	(1)	(2)	(3)	(4)	(5)	(6)
Bandwidth	10 km	11 km	12 km	13 km	14 km	15 km
Land fractionalization (1999)	13.067**	15.406**	17.998**	20.434**	23.280**	25.396**
	(5.946)	(7.162)	(8.199)	(9.296)	(10.429)	(11.298)
Baseline colleges (1999)	1.497***	1.606***	1.693***	1.793***	1.888***	1.950***
	(0.154)	(0.173)	(0.193)	(0.221)	(0.246)	(0.241)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	42,836	42,836	42,836	42,836	42,836	42,836
R-Squared	0.471	0.502	0.532	0.568	0.604	0.640

Notes: Robust standard errors clustered at the district level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is the change in the number of colleges within the stated bandwidth of a village between 1999 and 2007. Land fractionalization was measured at the district level in 1999. Baseline colleges (1999) are the number of colleges within the same bandwidth in 1999. Controls are the share of the Scheduled Caste population, the share of the Scheduled Tribe population, population density, elevation, terrain ruggedness, sandy-soil share, cropland share, night-time lights, and the share of non-farm employment. The Scheduled Caste and Scheduled Tribe shares, population density, and the non-farm employment share are sourced from the 2001 Census; night-time lights and cropland share from SHRUG; elevation and terrain ruggedness are computed in ArcGIS; and the sandy-soil share from the Soils of India database (NBSS & LUP, ICAR).

College Density by Year (15 km)

- Year-by-year fractionalization gradient within 15 km:

$$C_{v,t}^{15} = \alpha_t + \beta_t F_{d(v)} + X_v' \delta_t + \varepsilon_{v,t}, \qquad t = 1999, \ldots, 2007$$

  - In 1999, the coefficient is small and statistically insignificant (7.50), remaining insignificant through 2001.
  - From 2002 onward, the coefficient rises steadily and is statistically significant at the 10% level, reaching 47.53 in 2007, roughly six times its 1999 value.

	Number of colleges within 15 km								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Year	1999	2000	2001	2002	2003	2004	2005	2006	2007
Land fractionalization (1999)	7.502	9.584	10.813	12.726*	17.305*	22.840*	28.648*	35.777*	47.525*
	(4.972)	(5.811)	(6.433)	(7.300)	(9.868)	(12.708)	(16.003)	(20.687)	(25.419)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	42,836	42,836	42,836	42,836	42,836	42,836	42,836	42,836	42,836
R-Squared	0.179	0.183	0.177	0.169	0.172	0.172	0.179	0.186	0.204

Notes: Robust standard errors clustered at the district level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is the number of colleges within 15 km of a village in the year indicated by the column; each column is a separate cross-sectional regression. Land fractionalization was measured at the district level in 1999. Controls are the share of the Scheduled Caste population, the share of the Scheduled Tribe population, population density, elevation, terrain ruggedness, sandy-soil share, cropland share, night-time lights, and the share of non-farm employment. The Scheduled Caste and Scheduled Tribe shares, population density, and the non-farm employment share are sourced from the 2001 Census; night-time lights and cropland share from SHRUG; elevation and terrain ruggedness are computed in ArcGIS; and the sandy-soil share from the Soils of India database (NBSS & LUP, ICAR).

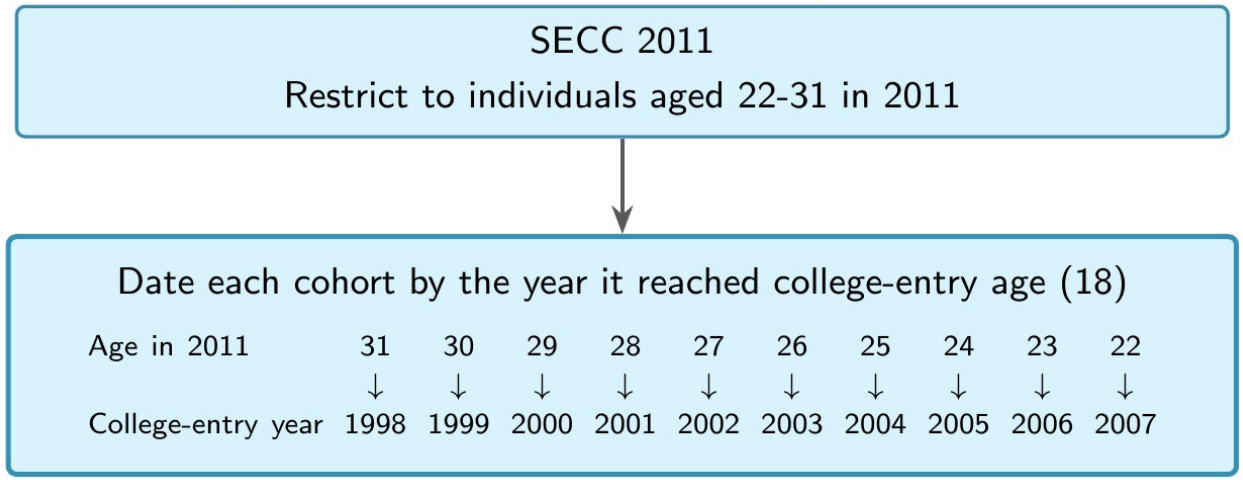
Conclusion

- The disproportionate expansion of colleges in Rajasthan increased college access and local tertiary attainment.
  - Results are robust to a range of several checks.
  - Both women and men experienced gains in tertiary attainment.
- College expansion was associated with pre-existing local land structure, particularly rural land fractionalization.

Thank You

Appendix

Construction of College-Entry Cohorts

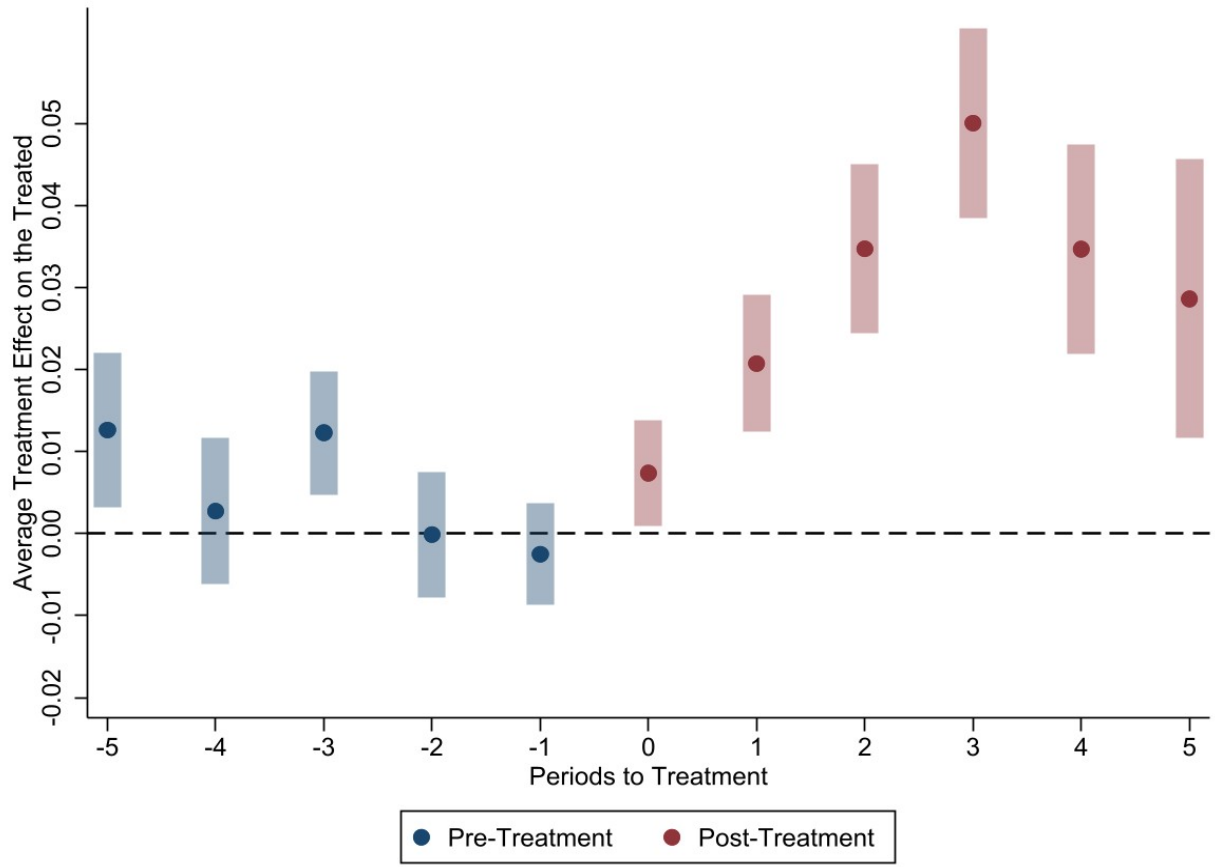


- College-entry year $=$ 2011 $-$ age in 2011 $+$ 18.
- Each village-cohort cell is assigned the level of college access prevailing in the corresponding college-entry year.

Back

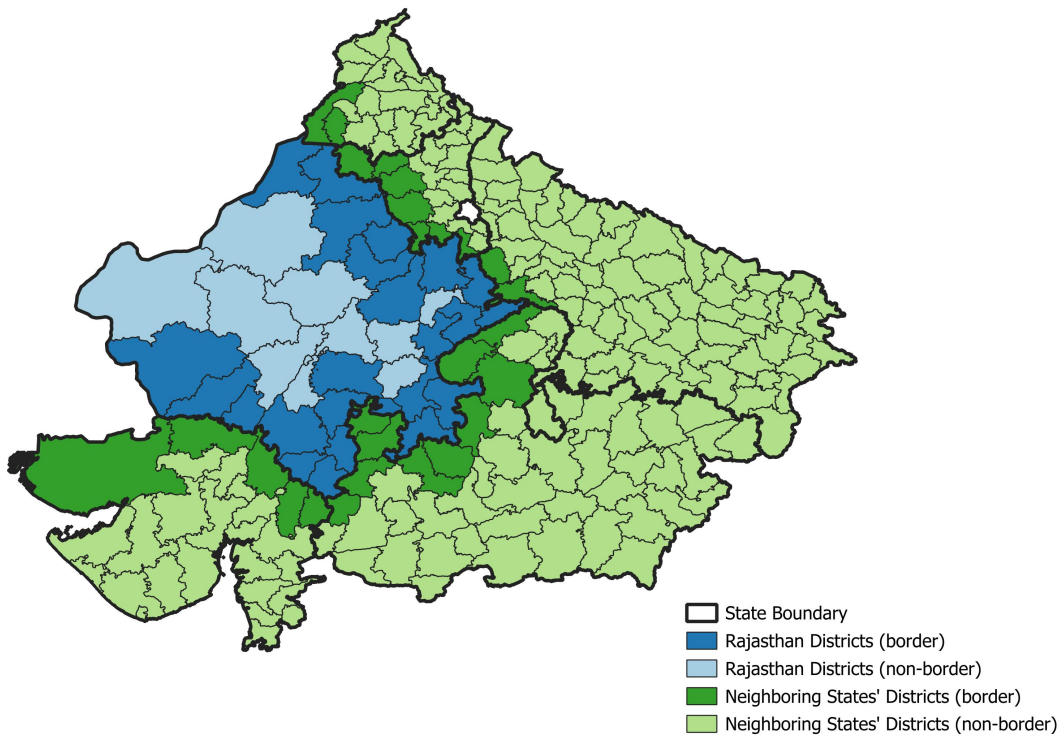
College Access and Local Night-Time Luminosity

- Outcome variable: $\ln(1 + \text{DMSP luminosity})$.
- Post-exposure: $\approx 0.7\%$ at $e = 0$ (marginal), 2.1% at $e = 1$, 3.5% at $e = 2$, peaking at 5.0% at $e = 3$, followed by 3.5% and 2.9%.
- Pre-exposure: estimates are near zero at $e = -4, -2, -1$, but are approximately 1.2%-1.3% and statistically significant at $e = -5, -3$.



College Expansion: The Contiguous Border District Sample

- Was the post-2003 expansion specific to Rajasthan, or did it reflect a national trend?
- Treated group: 22 border districts in Rajasthan. Control group: 26 adjacent districts in Gujarat (5), Haryana (7), Madhya Pradesh (10), Punjab (2), and Uttar Pradesh (2).



College Expansion: Event-Study Design

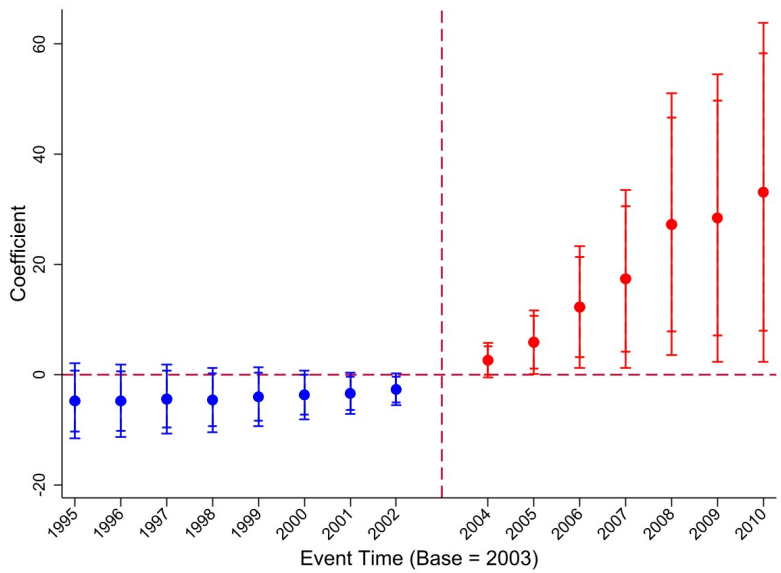
- District-year AISHE panel, 1995-2010:

$$Y_{dt} = \alpha_d + \lambda_t + \sum_{\substack{k=1995 \\ k \neq 2003}}^{2010} \beta_k \left(RJ_d \times \mathbb{1}[t = k] \right) + \varepsilon_{dt}$$

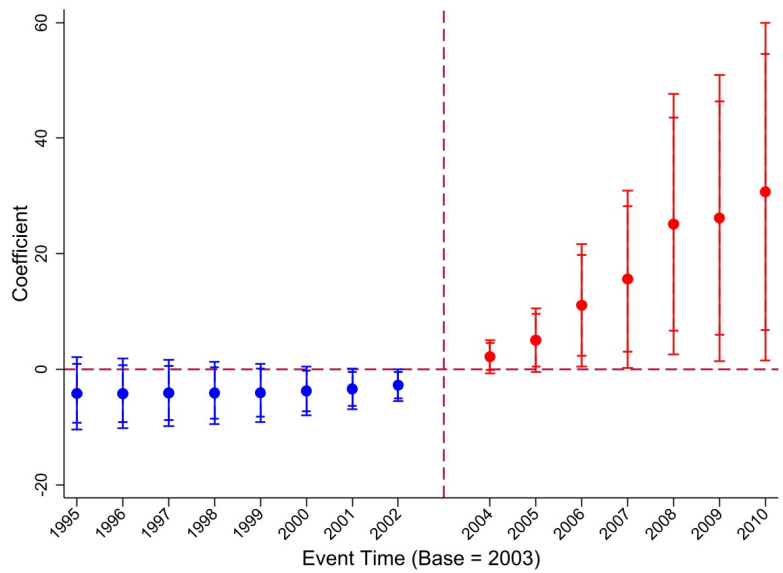
  - Y_{dt}: cumulative number of colleges established in district d up to year t.
  - $RJ_d = 1$ for Rajasthan border districts; α_d and λ_t denote district and year fixed effects, respectively.
  - Standard errors are clustered by district; 2003 is the omitted year.
  - β_k for $k \leq 2002$ tests for parallel pre-trends.

College Expansion: Results

- Before 2003: no discernible trend; confidence intervals include zero.
- From 2004 onward: positive and rising effects, statistically significant at the 5% level within one to two years; approximately 33 additional colleges per district by 2010.
- Private colleges follow an almost identical pattern, with approximately 31 additional colleges per district by 2010.



(a) Overall colleges



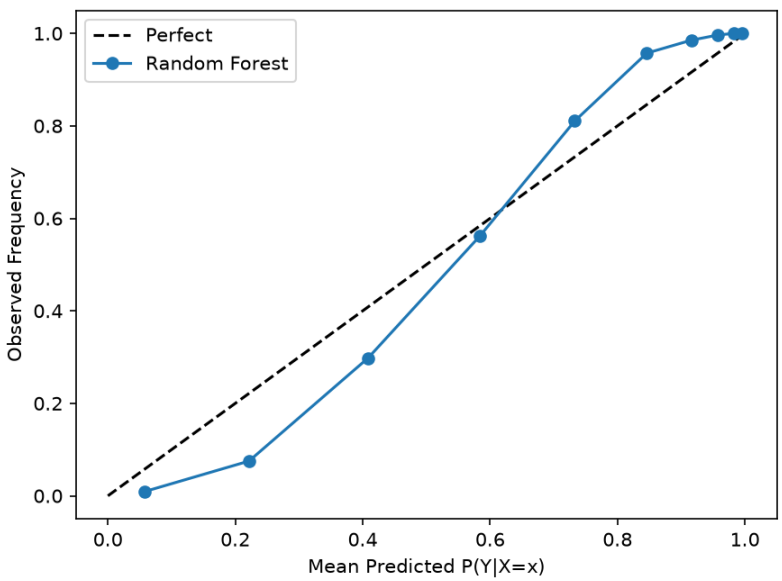
(b) Private colleges

Predictors of College Proximity: Setup

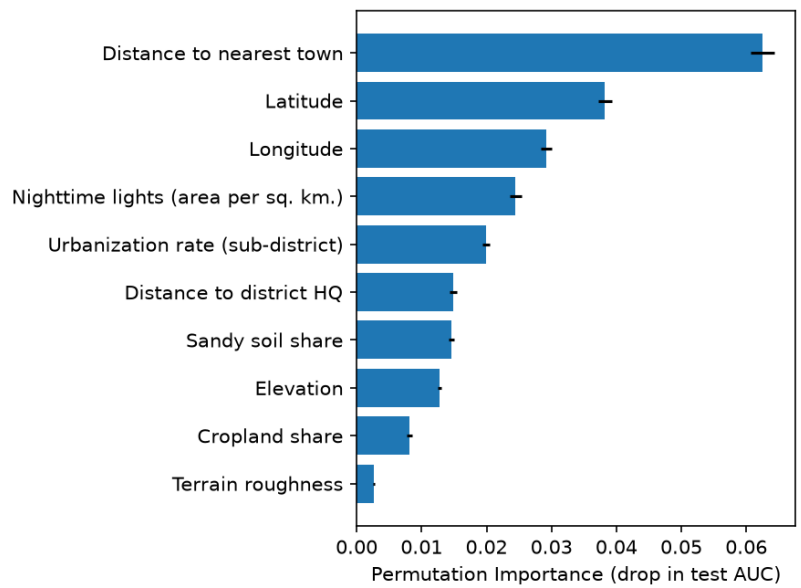
- Which village characteristics best predict proximity to a college?
  - Outcome: 1 if the village lies within 15 km of a college established by 2011.
  - 43,136 villages; 31 predictors:
    1. Geography and terrain: latitude, longitude, elevation, roughness, and sandy soil;
    2. Demographic and socioeconomic characteristics: population density, sex ratio, male and female literacy, SC/ST shares, and non-farm employment;
    3. Land use and economic activity: cropland and forest shares, night-time lights, and sub-district urbanization;
    4. Infrastructure and market access: schools, roads, highways, bus services, electricity, markets, banks, PHCs, distance to district HQ, and distance to the nearest statutory town.
- Random forest: 1,000 trees, $\sqrt{p}$ features per split, minimum leaf size of 5; stratified 25% hold-out and 5-fold cross-validation.

Predictors of College Proximity: Performance and Importance

- Held-out AUC = 0.962; 5-fold CV AUC = 0.964; precision-recall AUC = 0.982; Brier score = 0.079. Predictions lie close to the 45-degree line in the upper probability range, where most villages lie, but show some overprediction in the lowest bins.
- Permutation importance: distance to the nearest statutory town 26.3%, latitude 16.1%, longitude 12.3%, night-time lights 10.3%, sub-district urbanization 8.4%, and distance to district HQ 6.2%; jointly $\approx 80\%$.
- Sandy soil, elevation, cropland, and roughness account for a further $\approx 16\%$. Seven demographic and socioeconomic variables jointly account for 4.0% ($< 1\%$ each). Each infrastructure variable accounts for $\leq 0.1\%$.



(a) Calibration



(b) Permutation importance